Biotecnología médica III: VACUNAS

Valeria Tekiel
Introducción a la Biotecnología
UNSAM

1

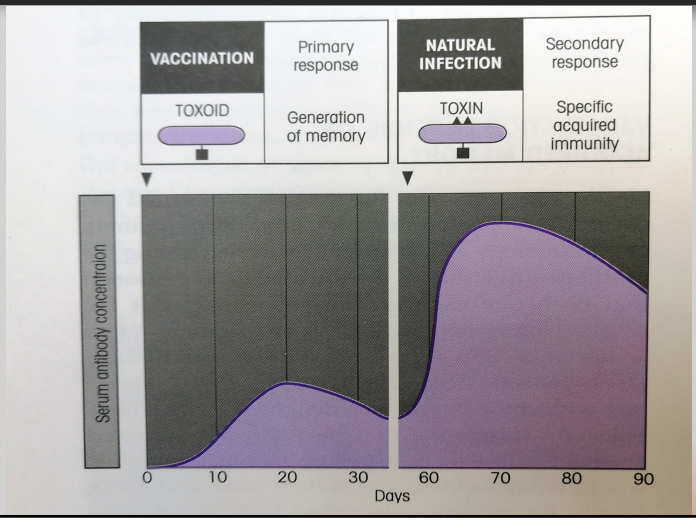
VACUNAS

Table 7.1 Annual cases in Canada from various diseases before and after the introduction of vaccines against the causative agents of the diseases

Disease	Annual no. of cases before vaccine was introduced	No. of cases in 2002
Polio	20,000	0
Diphtheria	9,000	0
Rubella	69,000	16
Mumps	52,000	197
Haemophilus influenzae type b infection	2,000	48
Whooping cough	25,000	2,557
Measles	300,000	7

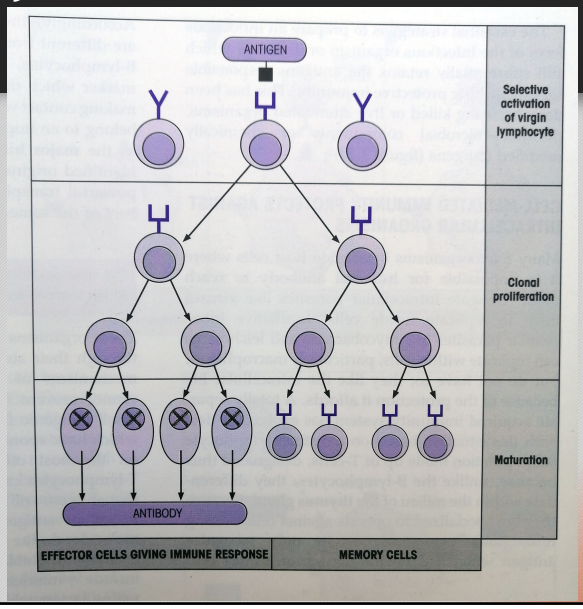
2

Respuesta inmune natural frente a infecciones e inducida por vacunación



3

Expansión clonal de linfocitos B: LBs secretores y LBs de memoria



4

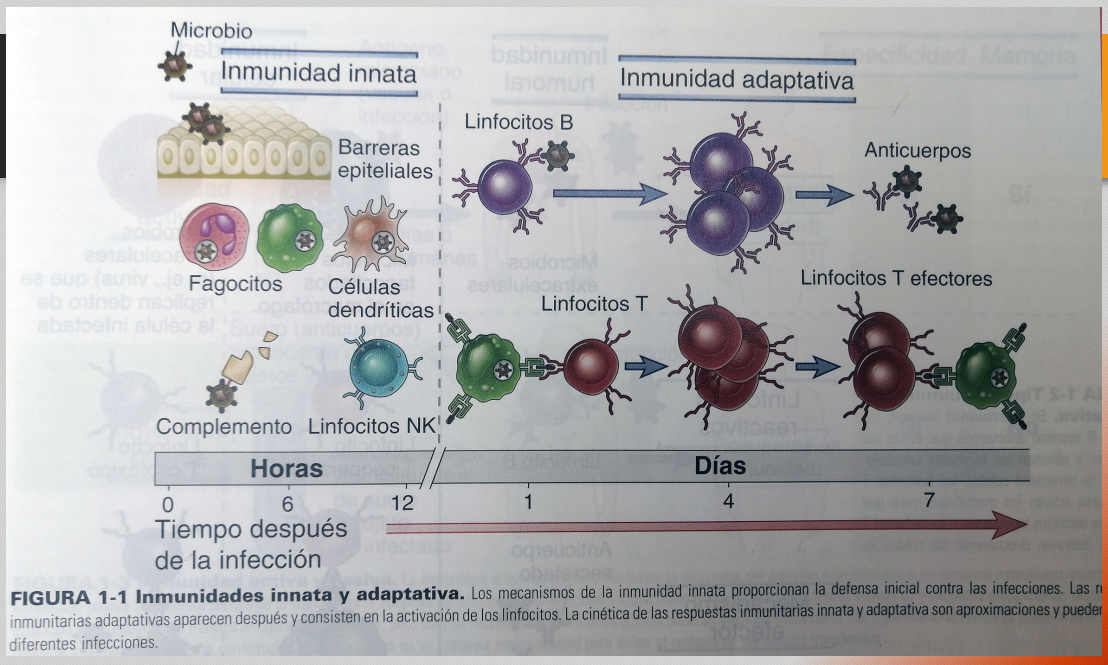
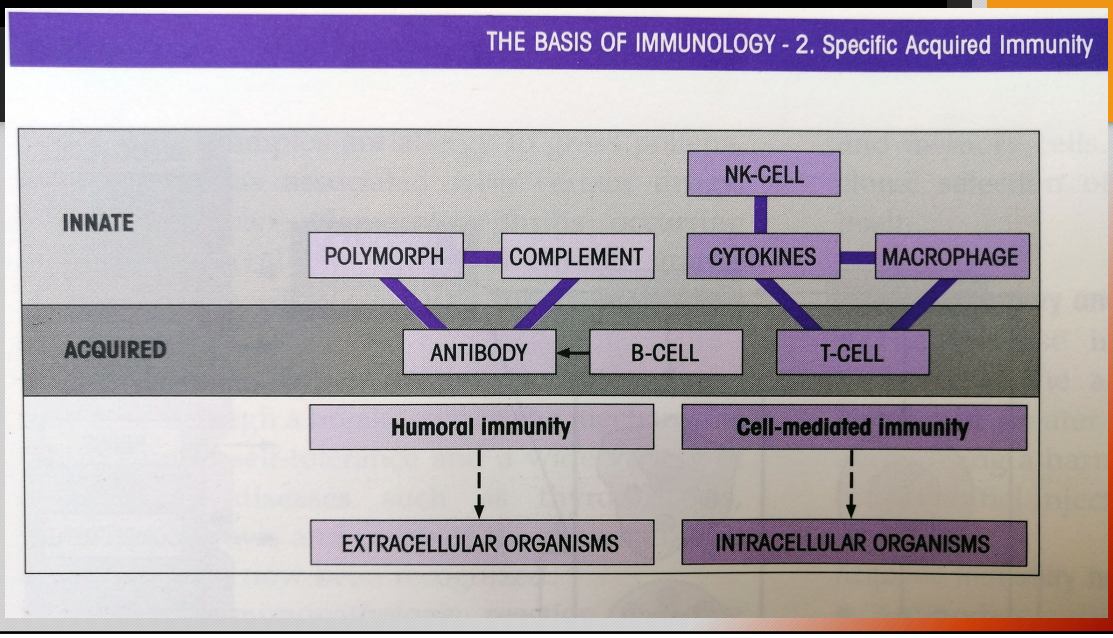


FIGURA 1-1 Inmunidades innata y adaptativa. Los mecanismos de la inmunidad innata proporcionan la defensa inicial contra las infecciones. Las r
inmunitarias adaptativas aparecen después y consisten en la activación de los linfocitos. La cinética de las respuestas inmunitarias innata y adaptativa son aproximaciones y pueden
diferentes infecciones.

5

Inmunidad innata y adaptativa



6

Inmunidad adaptativa

	Adaptativa
Características	
Especificidad	Frente a antígenos microbianos y no microbianos
Diversidad	Muy grande; los receptores se producen por recombinación somática de segmentos génicos
Memoria	Sí
Falta de reactividad frente a lo propio	Sí
Componentes	
Barreras celulares y químicas	Linfocitos en epitelio; anticuerpos secretados en superficies epiteliales
Proteínas sanguíneas	Anticuerpos
Células	Linfocitos

7

VACUNA

Preparación destinada a generar inmunidad específica contra una enfermedad o patógeno

VACUNACIÓN:
Exposición de un individuo a un microoganismo ó parte del mismo sin causar enfermedad, pero provocando una respuesta inmune que podrá bloquear o matar al microorganismo si la persona se infecta

8

TIPOS DE VACUNAS

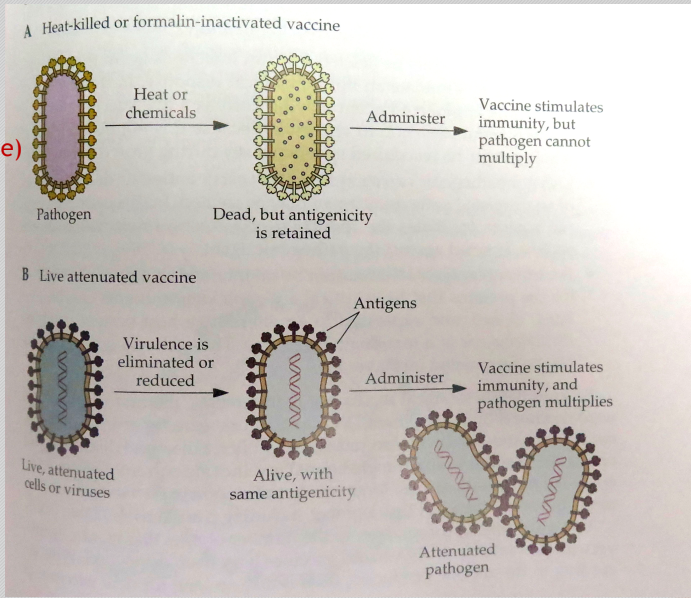
- TRADICIONAES (µorganismo en cultivo)
  - ATENUADAS
  - MUERTAS -INACTIVADAS
- TECNOLOGÍA ADN RECOMBINANTE → NUEVA GENERACIÓN VACUNAS
  - SUBUNIDADES
  - SISTEMAS VIVOS (atenuadas - carriers)
  - INMUNIZACIONES GENÉTICAS

9

VACUNAS TRADICIONALES

inactivadas
Polio (inyectable)
Sarampión

atenuadas
(vivas!)
Polio (oral)
sarampión
Paperas
Varicela
Rubeola
rotavirus
fiebre amarilla



Limitaciones
Cultivo µorg patógeno
Variación entre lotes
Seguridad-precauciones
Cadena frío y corta vida media

Reversión (atenuadas)
Inmunocomprometidos

No siempre funcionan ☹ (cq vacuna!)

10

VACUNAS A SUBUNIDADES

- PARTE del patógeno (gralmente proteínas o glicoprots exresadas en diferentes sistemas)
- RI contra epitopes específicos
- AG + **ADYUVANTE**
- **PROS**
  - **Proteína purificada → estable, segura, definida químicamente**
  - **Sin material extra (prot, ac nucleico) q genere RI indeseada**
- **CONS**
  - **Costos purificación**
  - **Baja antigenicidad**

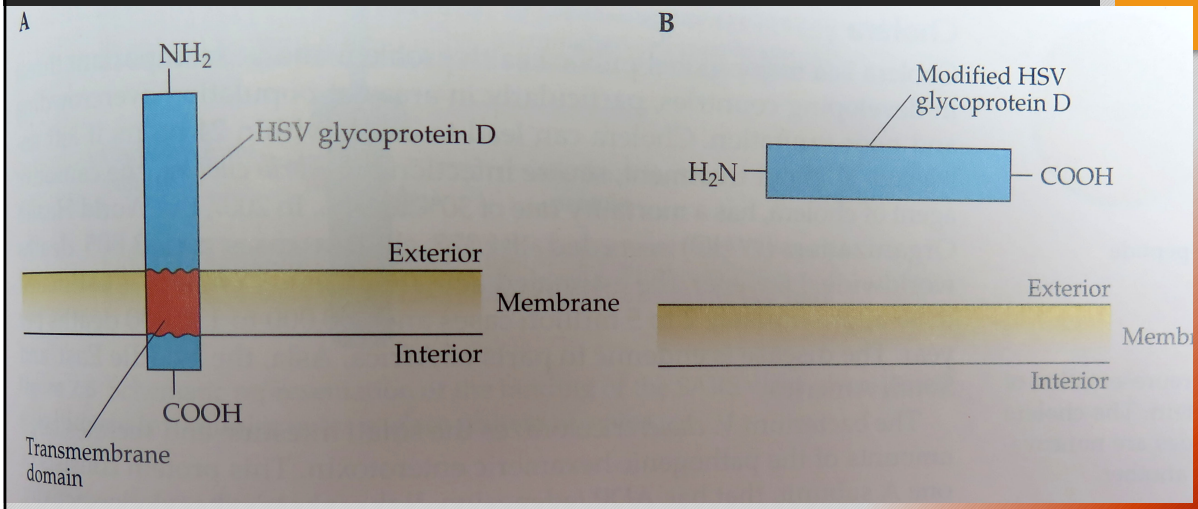
Subunidades *sensu stricto*, ej en el mercado:
HepatitisB
Meningococo (conjugada)
Neumococo
Influenza, etc

11

VACUNAS A SUBUNIDADES

EJ herpes simplex - glicoproteína D, en desarrollo

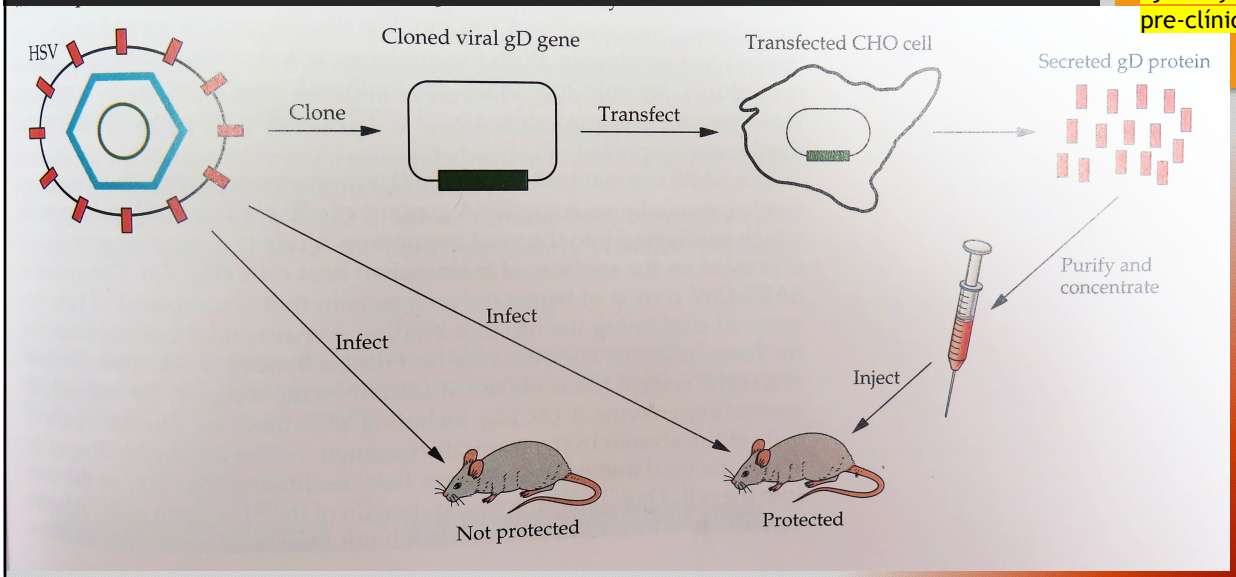
virus



12

VACUNAS A SUBUNIDADES EJ herpes simplex - glicoproteína D

ej ensayo pre-clínico



13

VACUNAS A SUBUNIDADES, TOXINA EJ cólera - toxina B

bacteria

Vacuna tradicional: *V. cholerae* muerto x fenol
protección moderada

Actual:
2 cepas inactivadas x calor + cepa inact x formalina + TOXINA B

oral, 2 dosis, 3 años inmunidad

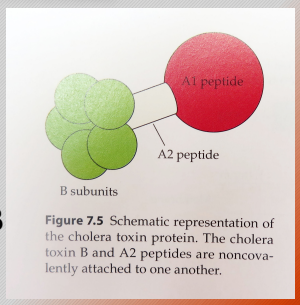
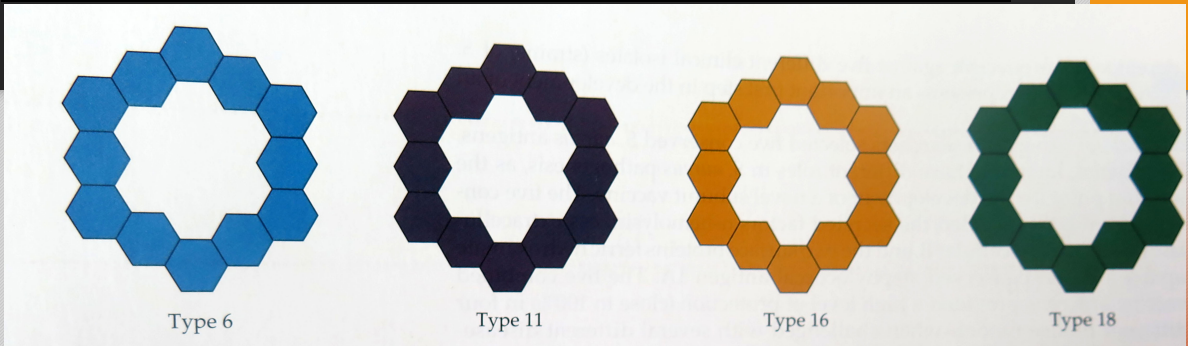


Figure 7.5 Schematic representation of the cholera toxin protein. The cholera toxin B and A2 peptides are noncovalently attached to one another.

ej toxinas: Tetanos, Difteria

14

VIRUS – LIKE PARTICLES (VLPs)
ej. HPV - serotipos



Moléculas que se ASEMEJAN a los virus → **PROTEÍNAS** derivadas de proteínas estructurales de virus

NO contienen material genético, por lo tanto no son infecciosos.

Se puede estimular su ensambaldo mediante la expresión de p del virus.

ej vacunas comerciales: HPV -expres prots en células de insecto ó levaduras

15

VACUNAS A SUBUNIDADES
SISTEMAS DE DELIVERY (administración)

ANTÍGENO + ADYUVANTE

VÍA INTRAMUSCULAR vs SUBCUTÁNEO

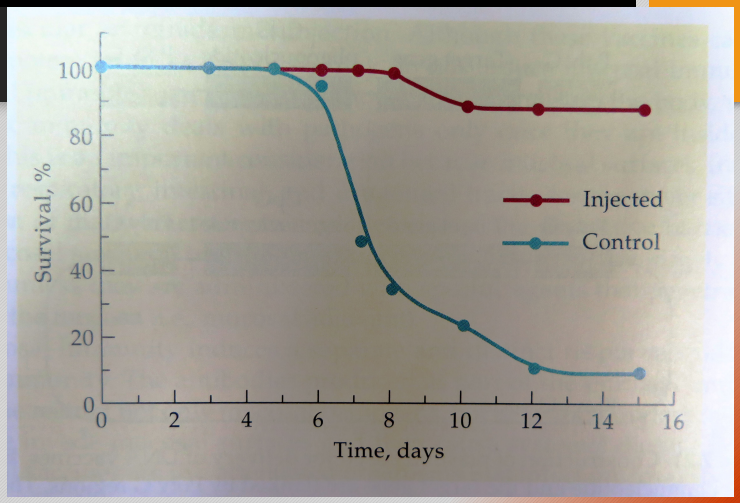
16

17

INMUNIZACIONES GENÉTICAS – ej VACUNAS A DNA

Table 7.4 Advantages of genetic immunization over conventional vaccines

Cultivation of dangerous agents is not required.
Since genetic immunization does not utilize any viral or bacterial strains, there is no chance that an attenuated strain will revert to virulence.
Since no organisms are used, attenuated organisms that may cause disease in young or immunocompromised animals are not a problem.
Approach is independent of whether the microorganism is difficult to grow or attenuate.
Production is inexpensive because protein does not need to be produced or purified.
Storage is inexpensive because of the stability of DNA.
One plasmid could encode several antigens/ vaccines, or several plasmids could be mixed together and administered at the same time.



Vacunación de ratones con ADN codificante para nucleoproteína de influenza A; Control: "vacunados" con plásmido vacío

18

VACUNAS A DNA – INMUNIZACIÓN GENÉTICA

Cómo?
Sistemas delivery
PROS

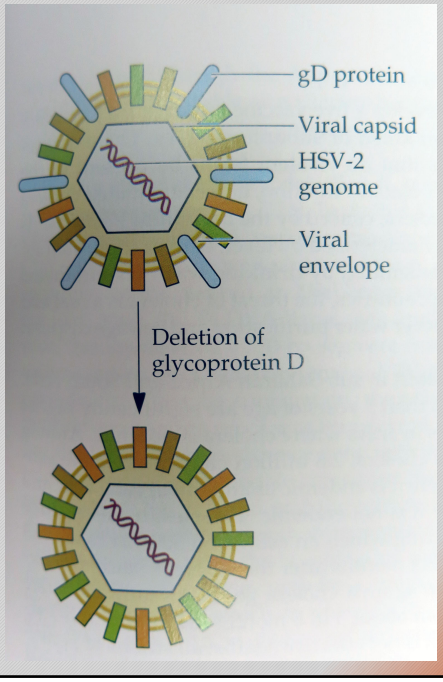
CONS

Table 7.5 Methods used to introduce DNA vaccines

Intramuscular injection
Microprojectile bombardment
Electroporation
Microparticles
Inside a bacterium
Inside a virus
Cationic liposomes
Nanoparticles
Together with cytokines
Together with adjuvants
Using a MIDGE vector

19

VACUNAS ATENUADAS de NUEVA GENERACIÓN



20

Tipos de vacunas: ventajas y desventajas

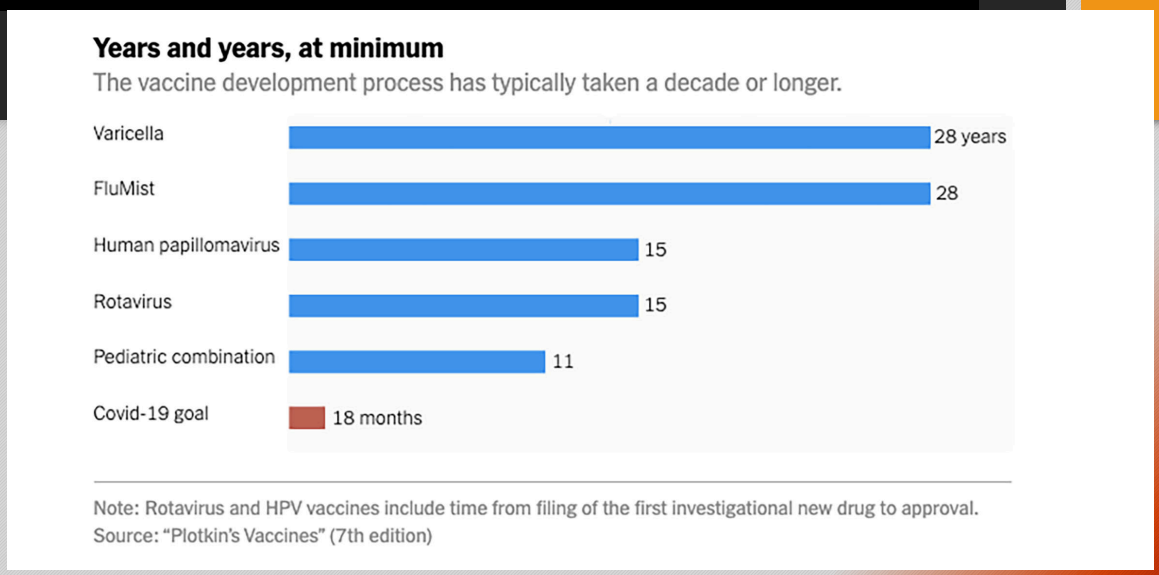
Table 1. Advantages and disadvantages of different vaccine strategies.

Vaccine strategy	Advantages	Disadvantages	References
Inactivated virus vaccines	Easy to prepare; safety; high-titer neutralizing antibodies	Potential inappropriate for highly immunosuppressed individuals	25
Attenuated virus vaccines	Rapid development; induce high immune responses	Phenotypic or genotypic reversion possible; can still cause some disease	25
Subunit vaccines	High safety; consistent production; can induce cellular and humoral immune responses; high-titer neutralizing antibodies	High cost; lower immunogenicity; require repeated doses and adjuvants	12,14
Viral vector vaccines	Safety; induces high cellular and humoral immune responses	Possibly present pre-existing immunity	12
DNA vaccines	Easier to design; high safety; high-titer neutralizing antibodies	Lower immune responses in humans; repeated doses may cause toxicity	23
mRNA vaccines	Easier to design; high degree of adaptability; induce strong immune responses	Highly unstable under physiological conditions	23

npj Vaccines (2020) 5:18 ; https://doi.org/10.1038/s41541-020-0170-0

21

Desarrollo de vacunas

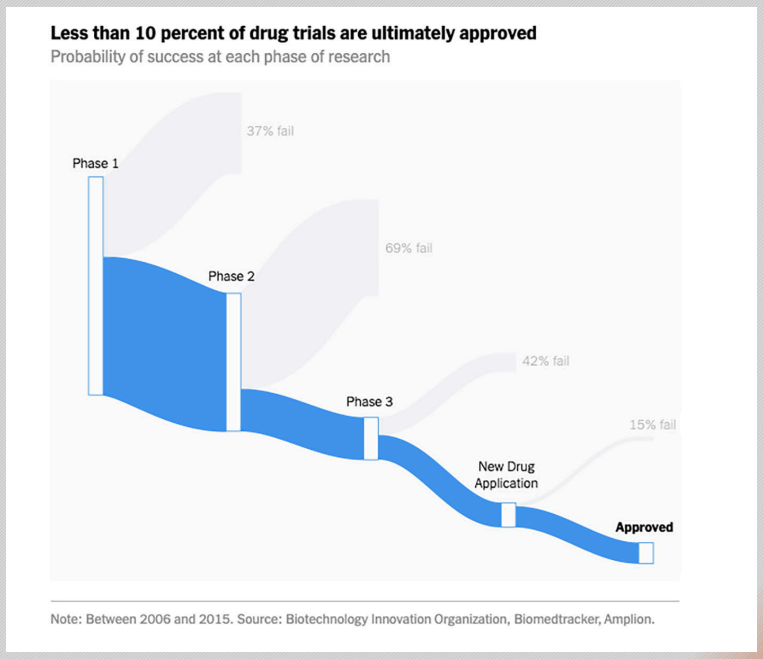


22

Etapas para desarrollo de vacunas

1. investigación académica
2. ensayos pre-clínicos (fase 0)
3. ensayos clínicos (fase 1, 2 , 3)
4. producción
5. aprobación final
6. distribución

23



24

Y.C. Kim et al.

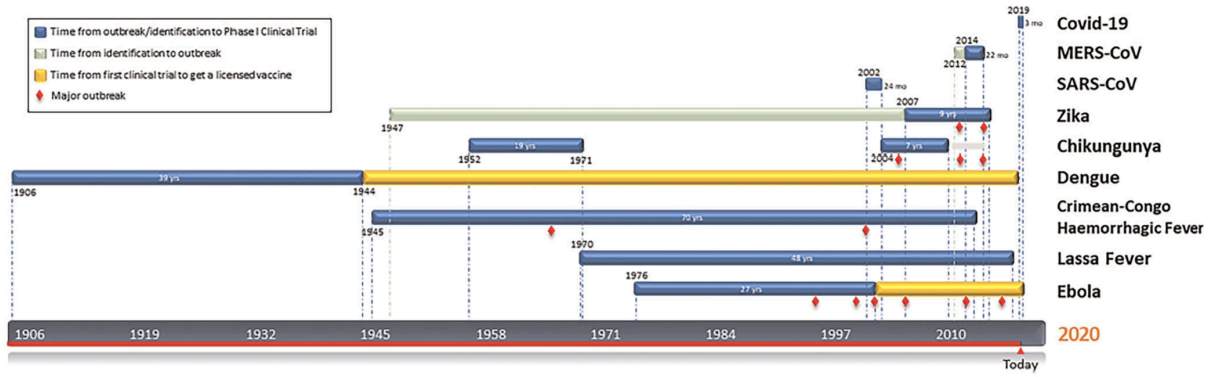


Fig. 1 New emerging diseases vaccine development timeline. Time since the first identification (green) of the virus in a patient or first major outbreak (blue) (and in some cases newly identification) to first Phase I Clinical Trial described. Orange bar represents the time it took to get a licensed Dengue/Ebola vaccine. Red diamonds indicate outbreaks.

- DengVaxia, aprobada 2019 en USA, sobre estructura vacuna fiebra amarilla, agregan prots serotipos 1,2, 3, 4
- **Qdenga o TAK-003 (lab TAKEDA)** APROBADA: 2023!! Virus Dengue-2 + DNA 3 serotipos

TAK-003 comprises 4 components (DENV1-4), with a live-attenuated DENV2 backbone, each encoding structural proteins premembrane (prM) and envelope (E) from DENV serotypes 1, 2, 3, and 4

otras vacunas, prots recomb en fase III npj Vaccines (2020) 5:34 ; https://doi.org/10.1038/s41541-020-0188-3

25

JOURNAL OF VIROLOGY, Nov. 2003, p. 11436–11447 Vol. 77, No. 21
0022-538X/03/$08.00+0 DOI: 10.1128/JVI.77.21.11436–11447.2003
Copyright © 2003, American Society for Microbiology. All Rights Reserved.

Dengue 2 PDK-53 Virus as a Chimeric Carrier for Tetravalent Dengue Vaccine Development

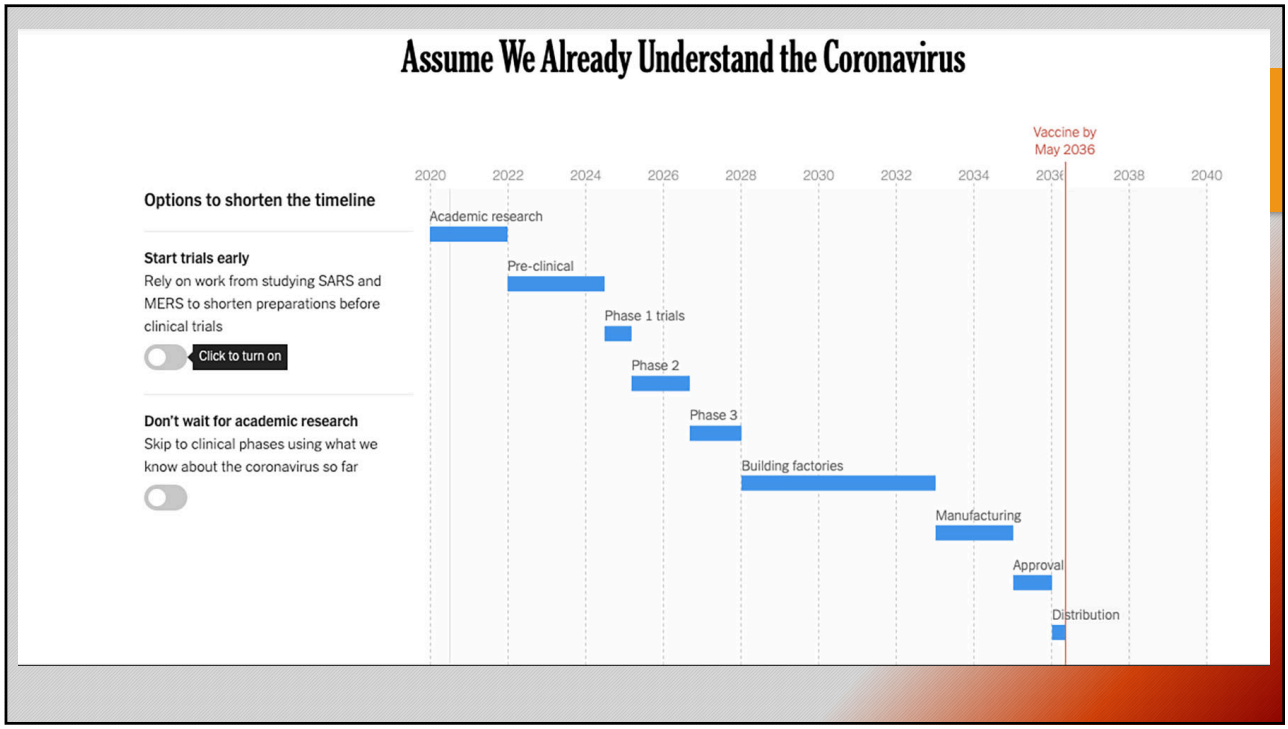
Claire Y.-H. Huang,[1]* Siritorn Butrapet,[1,2] Kiyotaka R. Tsuchiya,[1] Natth Bhamarapravati,[2] Duane J. Gubler,[1] and Richard M. Kinney[1]

Division of Vector-Borne Infectious Diseases, Centers for Disease Control and Prevention, Public Health Service, U.S. Department of Health and Human Services, Fort Collins, Colorado 80522,[1] and Center for Vaccine Development, Institute of Science and Technology for Development, Mahidol University at Salaya, Nakhonpathom 73170, Thailand[2]

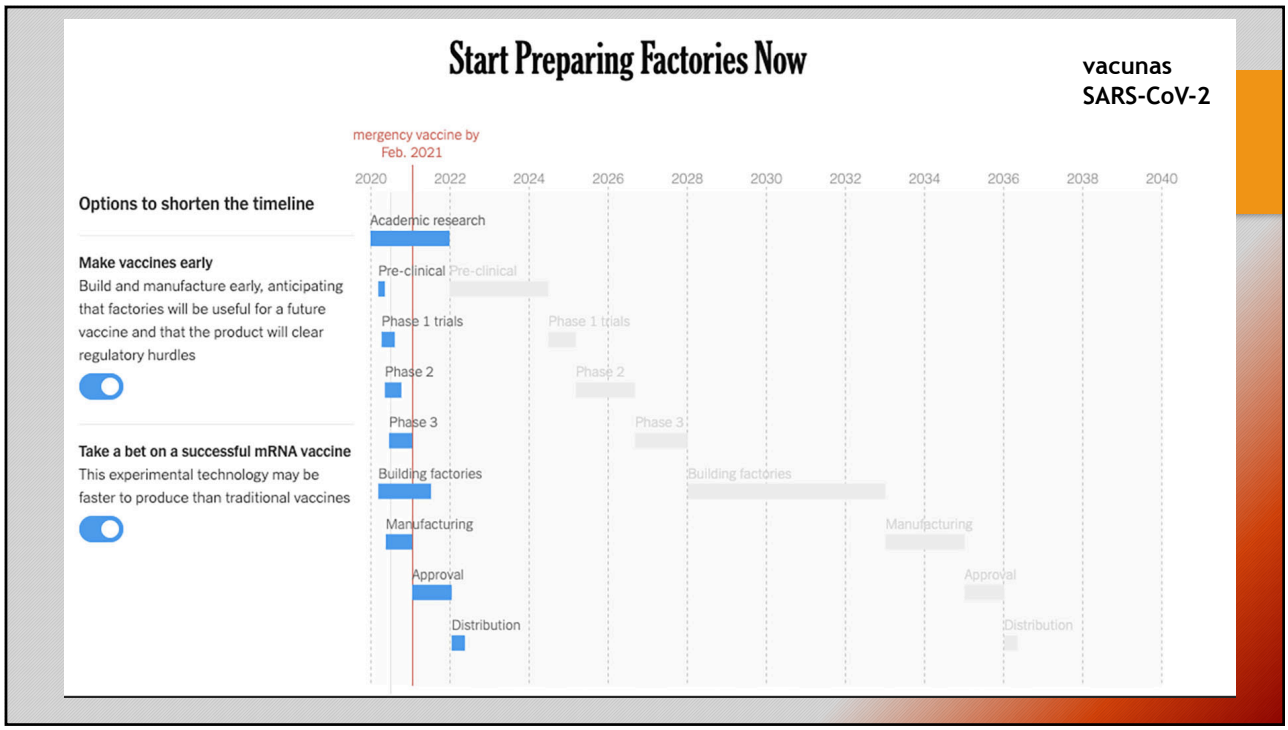
Received 4 June 2003/Accepted 24 July 2003

Attenuation markers of the candidate dengue 2 (D2) PDK-53 vaccine virus are encoded by mutations that reside outside of the structural gene region of the genome. We engineered nine dengue virus chimeras containing the premembrane (prM) and envelope (E) genes of wild-type D1 16007, D3 16562, or D4 1036 virus within the genetic backgrounds of wild-type D2 16681 virus and the two genetic variants (PDK53-E and PDK53-V) of the D2 PDK-53 vaccine virus. Expression of the heterologous prM-E genes in the genetic backgrounds of the two D2 PDK-53 variants, but not that of wild-type D2 16681 virus, resulted in chimeric viruses that retained PDK-53 characteristic phenotypic markers of attenuation, including small plaque size and temperature sensitivity in LLC-MK$_2$ cells, limited replication in C6/36 cells, and lack of neurovirulence in newborn ICR mice. Chimeric D2/1, D2/3, and D2/4 viruses replicated efficiently in Vero cells and were immunogenic in AG129 mice. Chimeric D2/1 viruses protected adult AG129 mice against lethal D1 virus challenge. Two tetravalent virus formulations, comprised of either PDK53-E- or PDK53-V-vectored viruses, elicited neutralizing antibody titers in mice against all four dengue serotypes. These antibody titers were similar to the titers elicited by monovalent immunizations, suggesting that viral interference did not occur in recipients of the tetravalent formulations. The results of this study demonstrate that the unique attenuation loci of D2 PDK-53 virus make it an attractive

26


Assume We Already Understand the Coronavirus
Vaccine by May 2036
2020
2022
2024
2026
2028
2030
2032
2034
2036
2038
2040
Options to shorten the timeline
Start trials early
Rely on work from studying SARS and MERS to shorten preparations before clinical trials
Click to turn on
Don't wait for academic research
Skip to clinical phases using what we know about the coronavirus so far
Academic research
Pre-clinical
Phase 1 trials
Phase 2
Phase 3
Building factories
Manufacturing
Approval
Distribution

27


Start Preparing Factories Now
vacunas SARS-CoV-2
mergency vaccine by Feb. 2021
2020
2022
2024
2026
2028
2030
2032
2034
2036
2038
2040
Options to shorten the timeline
Make vaccines early
Build and manufacture early, anticipating that factories will be useful for a future vaccine and that the product will clear regulatory hurdles
Take a bet on a successful mRNA vaccine
This experimental technology may be faster to produce than traditional vaccines
Academic research
Pre-clinical
Phase 1 trials
Phase 2
Phase 3
Building factories
Manufacturing
Approval
Distribution

28

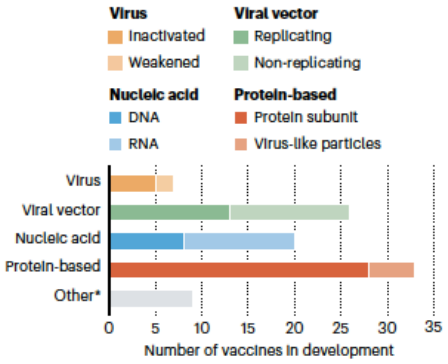
THE RACE FOR CORONAVIRUS VACCINES

By Ewen Callaway; design by Nik Spencer.

More than 90 vaccines are being developed against SARS-CoV-2 by research teams in companies and universities across the world. Researchers are trialling different technologies, some of which haven't been used in a licensed vaccine before. At least six groups have already begun injecting formulations into volunteers in safety trials; others have started testing in animals. *Nature*'s graphical guide explains each vaccine design.

AN ARRAY OF VACCINES

All vaccines aim to expose the body to an antigen that won't cause disease, but will provoke an immune response that can block or kill the virus if a person becomes infected. There are at least eight types being tried against the coronavirus, and they rely on different viruses or viral parts.



* Other efforts include testing whether existing vaccines against poliovirus or tuberculosis could help to fight SARS-CoV-2 by eliciting a general immune response (rather than specific adaptive immunity), or whether certain immune cells could be genetically modified to target the virus.

Nature | Vol 580 | 30 April 2020 | 577-580

29

desarrollo de vacunas en tiempos de pandemia

ej SARS-CoV-2

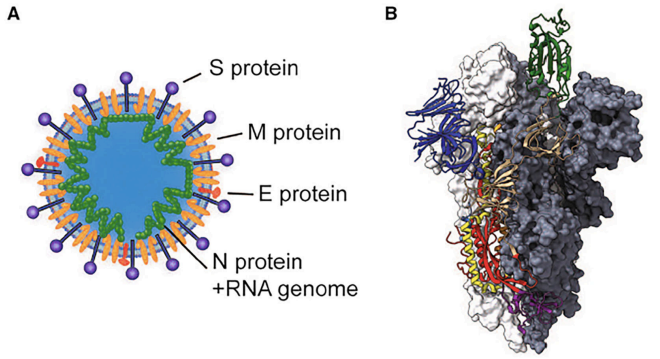


Figure 1. Structure of the SARS-CoV-2

(A) The SARS-CoV-2 is a spherical virus particle that incorporates three structural proteins into its lipid envelope (S, M, and E) and encapsidates a positive-stranded RNA genome in complex with the N protein. (B) The structure of the S protein trimer was solved using cryo-electron microscopy and is shown (PDB: 6VYB) (Wrapp et al., 2020). The RBD (shown in green) may existing in an open or closed (up or down) structural state; the RBD of one of the three S protomers within the trimer is shown in the up position.

Cell Host & Microbe, https://doi.org/10.1016/j.chom.2020.04.021

30

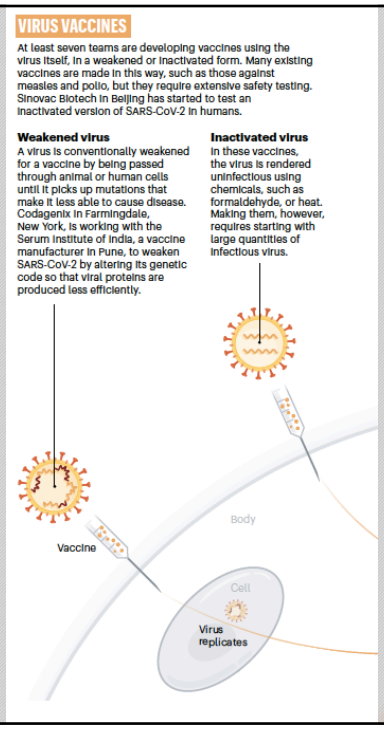
Table 1. Overview of Vaccine Production Platforms and Technologies for SARS-CoV-2

Platform	Target	Existing, Licensed Human Vaccines Using the Same Platform	Advantages	Disadvantages
RNA vaccines	S protein	No	No infectious virus needs to be handled, vaccines are typically immunogenic, rapid production possible.	Safety issues with reactogenicity have been reported.
DNA vaccines	S protein	No	No infectious virus needs to be handled, easy scale up, low production costs, high heat stability, tested in humans for SARS-CoV-1, rapid production possible.	Vaccine needs specific delivery devices to reach good immunogenicity.
Recombinant protein vaccines	S protein	Yes for baculovirus (influenza, HPV) and yeast expression (HBV, HPV)	No infectious virus needs to be handled, adjuvants can be used to increase immunogenicity.	Global production capacity might be limited. Antigen and/or epitope integrity needs to be confirmed. Yields need to be high enough.
Viral vector-based vaccines	S protein	Yes for VSV (Ervebo), but not for other viral vectored vaccines	No infectious virus needs to be handled, excellent preclinical and clinical data for many emerging viruses, including MERS-CoV.	Vector immunity might negatively affect vaccine effectiveness (depending on the vector chosen).
Live attenuated vaccines	Whole virion	Yes	Straightforward process used for several licensed human vaccines, existing infrastructure can be used.	Creating infectious clones for attenuated coronavirus vaccine seeds takes time because of large genome size. Safety testing will need to be extensive.
Inactivated vaccines	Whole virion	Yes	Straightforward process used for several licensed human vaccines, existing infrastructure can be used, has been tested in humans for SARS-CoV-1, adjuvants can be used to increase immunogenicity.	Large amounts of infectious virus need to be handled (could be mitigated by using an attenuated seed virus). Antigen and/or epitope integrity needs to be confirmed.

vacunas SARS-CoV-2

Cell Host & Microbe, https://doi.org/10.1016/j.chom.2020.04.021

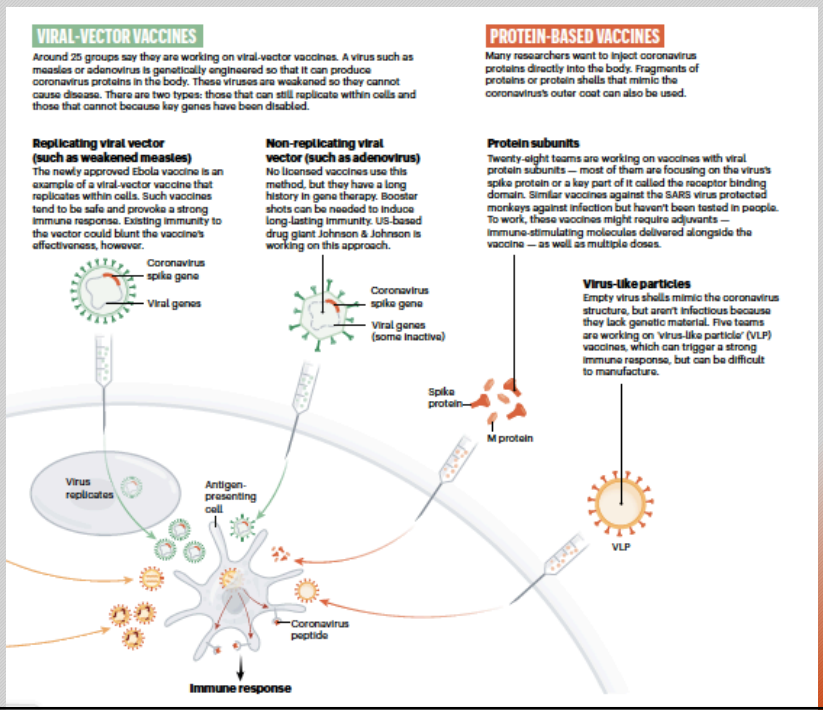
31



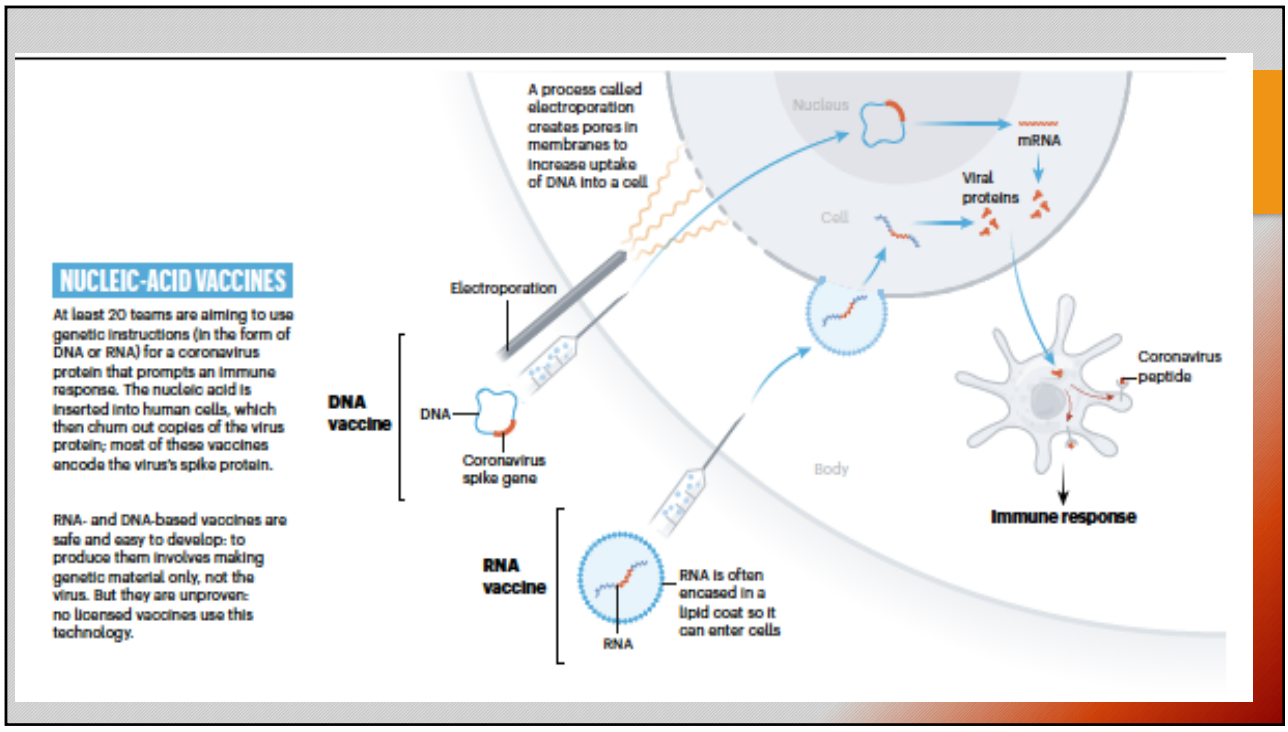
vacunas SARS-CoV-2

32

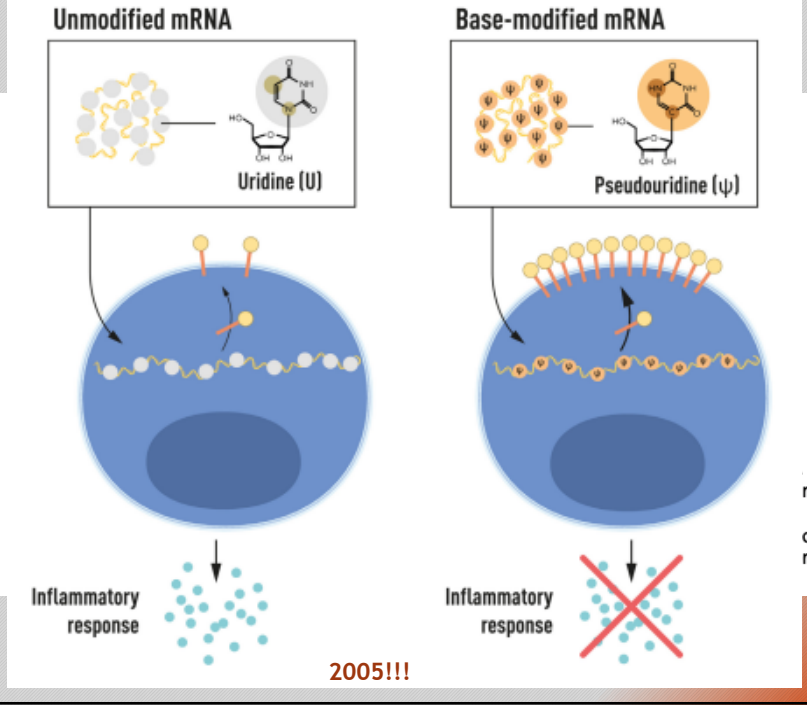
vacunas SARS-CoV-2



33



34



35

fin

Bibliografía
Molecular Biotechnology, (Glick&Patten), capítulo "Vaccines"

Essential Immunology, (Roitt, 8va edición), Cap 2: Specific acquired immunity
(o cualquier libro inmunología básica, capítulos introductorios

Lecturas adicionales
Subí algunos artículos en una carpetita
Tipos de vacunas: https://www.niaid.nih.gov/research/vaccine-types

36